MS 99 Executable business rules and reusable evidence contracts

Reconciled October 14 2026.

This milestone connects readable business rules to executable checks and separates agreement from evidence that can detect a wrong implementation. It also extracts common evidence-handling components so future adapters can reuse strict checks without inheriting an entire campaign implementation.

Delivered capability

The restricted rule evaluator produces signed receipts, offline replay, decision-point coverage, mutation results and HTML, JSON, DMN and scenario exports. Rule results retain their source anchors and mapping identities. A rule can agree with a fixture while its evidence remains weak or not assessed; those states are not presented as unqualified verification.

The later INTCALC public synthetic fixture exercises signed balances, account groups, rates and rounding boundaries. Its expected outputs come from the Python reference. The retained assessment finds discriminating evidence for disclosure-rate and monthly-interest only; other agreeing rules remain weak or not assessed. A surviving scale-increase mutant remains visible.

Shared evidence modules describe missing artifacts and authenticated available prefixes, compare explicitly selected tree and archive contents, cache reads only within a verification pass, and bind prebuilt shared bytes to an allowed consumer and saved outputs. They cannot sign decisions, launch a workload or manufacture missing evidence. Successful cleanup never converts a failed execution into a pass.

Evidence and remaining work

The rule records retain historical receipts, mutation ownership and the limited fields with observed diversity. Shared-module tests exercise tampering, omission, traversal, changed metadata and stale bindings. The build-once example records two host JVM consumers of identical shared classes; it is not a native database qualification.

Real Maintec outputs remain necessary for mainframe ground truth. The rule mutation adapter is specific to CardDemo rather than a general Java mutation engine. Adoption of the shared contracts by additional adapters and any live customer deployment require their own implementation and evidence.

Supporting records

- Executable business rules
- Evidence strength and remaining survivors
- Evidence completeness
- Content identity and per-pass caching
- Digest-bound build-once contract